AIRPORT EXECUTIVE ANALYSIS

Quiet-by-Design Standards for MWAA Terminals

A long-form argument for airport leaders

Transform Airports AI Council

August 12, 2026

DECISION SUPPORT • HUMAN REVIEW REQUIRED

> This decision-support document was generated with assistance from a multi-model AI research system. A named human decision owner remains responsible for verifying the evidence, judging local applicability, and approving any action.

The Loud Terminal Is the Unintelligible One

A communications-integrity standard for MWAA terminals — argued as safety engineering, delivered through the Design Manual.

Executive Summary

MWAA holds a committed capital construction program of approximately $6.99 billion at IAD and $2.39 billion at DCA — roughly $9 billion together — under the fifteen-year Airline Use and Lease Agreement effective January 1, 2025.[1] On July 29, 2026, the President, the U.S. Transportation Secretary, United's CEO, and MWAA jointly announced a Dulles transformation described at more than $20 billion, provisional pending Board action.[2] Every square foot of that work passes through the seven-volume MWAA Design Manual — "a mandatory guide with the force of law on the airport property."[3]

The recommendation is narrow: a tiered communications-integrity standard added to the Design Manual this capital cycle, issued by the Office of Engineering under its delegated standards-setting authority, taken to the Board as an information item, and escalated to a resolution only if airline consultation surfaces material objection. Reference application, subject to phase verification: the DCA Terminal 1 concourse replacement. Concourse E, opening this fall, carries the first commissioned intelligibility and ambient baseline under representative traffic.[4]

Keep "quiet-by-design" internal; externally, say "communications-integrity standard." After the January 29, 2025 midair collision and the NTSB's final report, no board member should sign a policy paraphrasable as "reduced safety broadcasts."[5] The standard is defensible only as the opposite: a floor that raises the probability the announcement that matters — the gate change, the shelter-in-place, the IROPS diversion — reaches the passenger who needs it. NFPA 72 requires emergency voice systems to achieve a Speech Transmission Index of at least 0.45 at 90% of measurement locations in each acoustically distinguishable space, average 0.50.[6] ACRP Research Report 175 recommends a minimum STI of 0.45 with 10 dB(A) signal-to-noise in pier-style departure lounges;[7] IEC 60268-16:2020 defines the Speech Transmission Index model and its measurement methods.[8] The reverberant, cross-broadcast terminal is the one that fails these numbers.

Three propositions carry the case. Acoustic treatment specified at design time costs a fraction of retrofit treatment at commercial benchmarks.[9] The enforcement vehicle already exists: the Design Manual binds consultants and contractors today, and extending it into tenant audio is the design objective, contingent on verification of the 2025 agreement's majority-in-interest provisions.[3] And the window is narrow: Moody's projects debt-service coverage compressing from 1.63× in 2024 to roughly 1.3× over five years as approximately $5.5 billion of new debt phases in,[10] which is why the amendment must be scoped to work still in programming. The Board question is not whether to fund a program. It is whether to write the paragraph.

The Loud Terminal Is the Unintelligible One

Start not at an airport but in a hospital intensive-care unit. In the alarm-fatigue studies the Joint Commission cites, monitored units log alarm counts that run to thousands a day across a floor — and clinicians had stopped hearing them.[11] The alarm that mattered sounded like the alarm that did not. The Joint Commission's response was not to reduce alarms in the name of patient comfort. It was National Patient Safety Goal NPSG.06.01.01, phased in 2014 and 2016: every accredited hospital had to inventory its alarms, rank them, govern them, and prove it.[12] Indiscriminate broadcast was itself the safety failure.

Picture the IAD international connecting bank at the late-afternoon peak. A gate change pages overhead. A TSA advisory. Two boarding calls at adjacent gates. A courtesy page. A concessions employee paging a colleague. In a reverberant holdroom shared with two adjacent gates, it is the alarm no one hears. IAD moved 29.01 million passengers in 2025, 10.53 million of them international — the fastest 2025 growth among the fifty largest US airports.[13] DCA ended the year down after the January collision, softened government-related travel, and a weeks-long federal shutdown. Whether the buildings were specified to let the announcement that matters reach those passengers is the open question.

MWAA is about to specify, from the studs up, an enormous run of new terminal space. The July 29 announcement covers more than 5 million square feet: Concourses C and D replaced, AeroTrain extended, mobile lounges retired, Saarinen's Main Terminal preserved.[2] It does not mention a single acoustic or public-address specification. The $20 billion-plus figure is provisional; the committed agreement figures are firm.[1] The anchor tenant has noticed the price: Cranky Flier described United's CEO as visibly reluctant and cited an airline briefing pegging fully loaded IAD transformation cost per enplanement at $90.64, against $12.88 today.[14] That reluctance strengthens the Design Manual case. A standard that lives in the Manual survives program shrinkage, re-scoping, or a change of administration. A case built on program durability does not.

The obvious objection first. After the NTSB's final report on the January 29, 2025 midair collision, no board member wants a policy paraphrased as "airport cuts safety announcements."[5] That instinct is correct — and it is the argument. The silent-airport literature equates quietness with subtraction. The evidence supports the inverse. NFPA 72 sets an intelligibility floor for emergency voice systems, scoped to acoustically distinguishable spaces, because a terminal is not one acoustic environment.[6] You fail that floor by being loud, reverberant, and cross-broadcast: precisely the condition an unspecified terminal defaults to. The louder the terminal, the less intelligible the emergency announcement. Every acoustic and PA-zoning requirement in the recommended standard flows from that inversion.

The peer set usually cited is a category error. London City has run a silent policy since 2008.[15] Helsinki-Vantaa, silent since 2015, and Copenhagen moved information to screens and apps.[16] All three are small, point-to-point operations — quieter to begin with. Schiphol is the real comparison: a large transfer hub whose reduced-announcement regime rests on screen density and an app-carrying passenger base.[16] None of the four has published measured intelligibility, missed-boarding, or channel-redundancy data. They are policy analogues, not design precedents. MWAA does not need to import their brand. It needs to out-engineer them.

SFO is the closest North American precedent, and it shows both the model and the failure mode. SFO reports its Quiet Airport program cut individual paging from 492 to 261 occurrences per day and cumulative announcement time from 145 to 58 minutes.[17] Every figure is SFO's own. The same record holds documented passenger complaints and missed-flight anecdotes.[18] Borrow the operating model — zoned delivery, no cross-terminal final calls, gate-area retention — and treat the outcome numbers as SFO's own accounting, not as facts. Announcement reduction without intelligibility engineering and channel redundancy is subtraction, not design.

The recommendation must pass one test above all: the people who most need to hear. 28 CFR 35.160 obligates MWAA to ensure effective communication with people with disabilities.[19] The 2010 ADA Standards §219 and §706 require assistive-listening systems in each assembly area where audible communication is integral to use.[20] Under 14 CFR Part 382, passengers who need visual or hearing assistance must receive prompt access to the same trip information other passengers get at the gate.[21] ACRP Research Report 239 treats digital-first substitution as partial, not sufficient, for travelers with disabilities and older adults.[22] A regime that reduces broadcast without specifying redundant visual, digital, staffed, and hearing-loop channels compounds exclusion. A standard that mandates those redundancies binds MWAA harder than current practice, not softer. One schedule note: the Saarinen Main Terminal has held National Register of Historic Places status since 1978, and Section 106 review adds months to any adjacent scope[23] — reason to keep the amendment's first applications outside the historic-district envelope.

Governance is the technology. The newer automated announcement platforms can be specified in Design Manual terms: centralized management, replay and audit logs of every broadcast, elimination of manual-initiation errors, templated multi-language delivery.[24] This is the Joint Commission move applied to airports — inventory, rank, govern, log — run by a standing governance committee reporting annually to the Board. Enforcement matters as much as the platform: annual acoustic inspections keyed to lease renewal, cure notices, lease-management escalation only for pattern noncompliance. Without those mechanics, airlines and concessionaires read the standard as paper.

Which raises the tenant question. At preferential-use gates, the podium PA is effectively airline-owned. United controls 67% of IAD passenger traffic and operates 50 jetbridge-equipped gates; American holds roughly 53% of DCA market share.[25] A standard that cannot reach tenant audio governs half the soundscape. Nobody claims today's Design Manual reaches the podium microphone. The argument is that the amendment, once verified against the use-and-lease agreement's majority-in-interest provisions, should be written to reach it — with carriers briefed early, a formal comment window, and MWAA's cost-recovery position in writing before amendment language enters any scope document.

The financial case is rework avoidance, not new capital. Commercial pricing guides put new-construction acoustic ceiling treatment at roughly \$5–\$15 per square foot and retrofit panel treatment at roughly \$18–\$45.[9] Non-airport, medium-confidence benchmarks — but they point one direction: specify at design time. The window is narrow and directional: Moody's projects debt-service coverage compressing from 1.63× in 2024 to roughly 1.3× over five years as approximately \$5.5 billion of new debt phases in.[10] Added specification cost inside programs already in design cuts against coverage, which is why the amendment must be scoped to work still in programming.

The standard writes itself as a two-tier document. Mandatory at both airports: an acoustically-distinguishable-space map as a design deliverable; intelligibility floors per NFPA 72 and ACRP 175; measurement per IEC 60268-16:2020, commissioned under representative traffic with a cure period — the standard itself does not cover fluctuating background noise, which is exactly why an empty-building reading is not a commissioning result;[26] accessibility floors per 28 CFR 35.160, the ADA Standards, and Part 382; a hearing conservation program per 29 CFR 1910.95, occupational noise exposure;[27] audit-logged announcement governance; and concessionaire televisions and gate-lounge audio kept inside scope, with thresholds set in the amendment. Then the paragraph most silent-airport writing omits: an IROPS override. During declared irregular operations, the Airport Duty Manager holds named authority to exceed announcement-density templates; every deviation is logged and reconciled afterward. A standard that does not survive the first Wednesday afternoon of gate holds will not be believed for the Wednesday it was written for.

Exercise it first where friction is lowest. Subject to project phase — not verified in this research pass — the DCA Terminal 1 concourse replacement is the natural choice: smaller, less politicized, less encumbered. If Terminal 1 has passed scope definition, the reference application shifts to the earliest IAD C/D element still in programming.

Either way, Concourse E — approximately 435,000 square feet, fourteen gates, opening this fall — carries the first commissioned intelligibility and ambient baseline under representative traffic.[4] That baseline becomes the number every future MWAA facility is measured against.

The strongest counter-case: NFPA 72, the ADA Standards, and ACRP 175 already set the floors that matter. Half-right. NFPA 72 binds *emergency* voice; it does not require acoustic-space mapping as a general design deliverable, does not govern routine PA, does not compel governance of manual-initiation error. The ADA Standards set receiver counts, not intelligibility floors for daily announcements. ACRP 175 is guidance, not code. The amendment's marginal content — the space map at each design milestone, representative-traffic commissioning, audit-logged governance, channel-redundancy floors, the IROPS override, tenant-audio boundaries — is exactly what the existing floors do not require. The premature-specification objection is more serious: rigid overlays written before program-brief maturity drive change orders, and hard commissioning gates on the critical path in this coverage window would be a bond-rating conversation. Hence verification-with-cure-period commissioning and the space map as a milestone deliverable rather than an occupancy condition.

Return to the ICU. The Joint Commission did not tell hospitals to reduce alarms. It told them to prove they knew which ones mattered, that those would be heard, and that the machinery for governing the rest was on the record.[12] A decade later the ICUs are not silent. They are audible in the right places, at the right times, to the right people. That is what a communications-integrity standard buys an airport. The Board question is not whether to fund a program. It is whether to write the paragraph.

Notes

[1] MWAA, "Metropolitan Washington Airports Authority Completes Negotiations for New Use and Lease Agreement with Airlines" (2024), mwaa.com: 15-year agreement effective January 1, 2025; capital construction program of approximately $6.99 billion at IAD and $2.39 billion at DCA — approximately $9 billion together; airline-funded debt-service coverage to at least a 140-percent target ratio. Majority-in-interest thresholds and cost-recovery categorization not verified from primary agreement text.

[2] July 29, 2026 joint announcement by the President, the U.S. Transportation Secretary, United, and MWAA, as reported in "Inside the $20B+ Plan to Remake Washington Dulles International Airport" (travelagentcentral.com) and *Simple Flying*, "Goodbye Mobile Lounges: Here Comes Washington Dulles' $20+ Billion Revival": more than 5 million square feet of new or refurbished facilities, Concourse C/D replacement, AeroTrain extension, mobile-lounge retirement, preservation of the Saarinen Main Terminal. The $20 billion-plus figure is provisional pending Board action and airline-consent documents.

[3] MWAA Office of Engineering, *Design Manual* (seven volumes, annually updated), mwaa.com; the quoted characterization appears in MWAA's own description. The Manual's current acoustic and PA content, and the reach of tenant technology standards into airline holdroom audio, were not verified from primary Manual text in this research pass.

[4] MWAA, Concourse E project disclosures (flydulles.com): approximately 435,000 square feet, 14 gates, initial phase opening fall 2026. Verified against the project page August 11, 2026.

[5] National Transportation Safety Board, Aircraft Accident Report NTSB/AIR-26/02, *Midair Collision over the Potomac River — PSA Airlines Flight 5342* (final, 2026, ntsb.gov). The January 29, 2025 collision and its 67 fatalities were documented in contemporaneous coverage: CNN, "DCA midair collision first responders, others remembered by victims' families as heroes" (cnn.com).

[6] NFPA 72, *National Fire Alarm and Signaling Code*, emergency communications systems intelligibility criteria: STI of at least 0.45 at 90% of measurement locations within each acoustically distinguishable space, average STI of at least 0.50. Primary code is paywalled; figures verified against technical secondary sources reproducing the code language (commercialintegrator.com, "Mass Notification: Intelligibility is NOT Optional") and corroborated across ten-plus records in the research base.

[7] Airport Cooperative Research Program, *ACRP Research Report 175: Improving Intelligibility of Airport Terminal Public Address Systems*, Transportation Research Board (2017): recommends a minimum STI of 0.45 and a signal-to-noise ratio of 10 dB(A) for public-address broadcasts in pier-style departure lounges (nap.nationalacademies.org/read/24839).

[8] International Electrotechnical Commission, IEC 60268-16:2020, *Sound system equipment — Part 16: Objective rating of speech intelligibility by speech transmission index* (webstore.iec.ch/en/publication/26771): defines the STI model, test signals, and measurement methods. Standard is paywalled; MWAA specifications can require compliance by reference.

[9] Commercial pricing guides: new-construction acoustic ceiling installation $5–$15 per square foot (designtransitionstudio.com, "Acoustical Ceiling Cost Per Square Foot: Price Guide 2026," which also quotes lower typical bands and premium tiers to $22); retrofit acoustic panel treatment $18–$45 per square foot (acousticmod.com, "Acoustic Retrofit vs. New Construction: Cost & Time Guide," as captured in research; page since relocated). Non-airport, medium-confidence directional benchmarks; airside premiums would raise both ranges.

[10] Moody's projection as reported in *The Bond Buyer*, "MWAA goes to market" (2025, bondbuyer.com): net revenue debt-service coverage declining from 1.63× in 2024 to around 1.3× over the next five years; approximately $5.5 billion of new debt planned 2025–2028; adjusted debt per O&D enplanement peaking near $400 in 2028 versus $223 in 2024. Underlying rating-agency reports are the primary citation-grade source.

[11] *Respiratory Therapy Magazine*, "Alarm Fatigue in Adult ICU" (respiratory-therapy.com): reviews the alarm-fatigue literature underlying clinical alarm-management policy, including a large academic medical center logging 74,535 alarms across eight units in one week. Reported per-bed figures vary by study; used as illustrative magnitude, not a universal baseline.

[12] The Joint Commission, National Patient Safety Goal NPSG.06.01.01, *Clinical Alarm Management*, effective January 1, 2014 (phase one), January 1, 2016 (phase two).

[13] MWAA 2025 traffic reporting: approximately 29.01 million passengers at IAD, 10.53 million of them international, the highest 2025 growth among the fifty largest U.S. airports. As reported in *Simple Flying*, "How United Is Quietly Rebuilding Washington Dulles Into Its Ultimate Hub" (simpleflying.com) and regional coverage (*FFXnow*, July 2, 2026).

[14] Brett Snyder ("Cranky Flier"), "The Plan for Dulles is Ambitious and Ridiculous… and It Won't Happen" (August 4, 2026, crankyflier.com): describes United CEO Scott Kirby's visible reluctance and cites an airline briefing at fully loaded IAD transformation cost per enplanement of approximately $90.64 versus $12.88 today. Attributed as analyst commentary; briefing methodology not independently verified.

[15] London City Airport, silent-airport policy implemented 2008; no flight or gate announcements since August 18, 2016, except essential and emergency (londoncityairport.com).

[16] Helsinki-Vantaa (Finavia, silent regime since 2015, gate-area and emergency overrides retained); Copenhagen Airport operator-published passenger-communication policy; Amsterdam Schiphol, operator-described "Silent Airport" model with information on screens and airport app. No published intelligibility or missed-boarding measurement identified for any of these airports in this research pass.

[17] San Francisco International Airport, Quiet Airport program outcomes as reported in trade press citing SFO: individual paging occurrences 492 to 261 per day; cumulative paging duration 145 to 58 minutes per day. Sources: *International Airport Review*, "San Francisco Airport continues work on Quiet Airport programme" (2023); *View from the Wing* (viewfromthewing.com). All figures SFO-reported; measurement methodology not published.

[18] *Simple Flying*, "Why This Airport's 'Quiet Policy' Is Frustrating Some Passengers" (simpleflying.com): contemporaneous accounts documenting missed-flight incidents and confusion attributed to SFO's reduced-announcement practice.

[19] 28 C.F.R. § 35.160 (Title II of the Americans with Disabilities Act): effective communication; auxiliary aids and services. See "ADA Requirements: Effective Communication" (ada.gov).

[20] U.S. Department of Justice, *2010 ADA Standards for Accessible Design*, §§ 219 and 706: assistive-listening systems required in assembly areas where audible communication is integral to use.

[21] 14 C.F.R. Part 382 (Air Carrier Access Act implementing regulations, ecfr.gov): carriers must make facilities and services accessible, including prompt access for passengers needing visual or hearing assistance to the same gate information provided to other passengers.

[22] Airport Cooperative Research Program, *ACRP Research Report 239: Assessing Airport Programs for Travelers with Disabilities and Older Adults* (Transportation Research Board, 2023): treats digital and app channels as complements rather than substitutes for staffed and audio communication.

[23] "Dulles International Airport Main Terminal" (en.wikipedia.org): the Eero Saarinen Main Terminal's National Register of Historic Places status dates to 1978. The FAA/ACHP record treats the terminal area as Register-eligible historic property, which attaches Section 106 consultation obligations to adjacent work.

[24] ACI-NA, "The Data-Driven Future of Automated Airport Announcements" (February 27, 2026, airportscouncil.org): describes automated PA platform capabilities including centralized management, audit-logged broadcast, elimination of manual-initiation error, and multi-language templating. Industry advocacy piece; directional rather than measurement-grade.

[25] United share and gates: *Simple Flying*, "How United Is Quietly Rebuilding Washington Dulles Into Its Ultimate Hub" (simpleflying.com): United controls 67% of IAD passenger traffic and operates 50 jetbridge-equipped gates in Concourses C and D; verified August 11, 2026. American share: *Simple Flying*, "American Airlines' Most Popular Routes From Its Washington DC Hub" (March 2025): approximately 53% DCA market share at end-2024. Directional trade-press figures, not primary MWAA rate-setting disclosure.

[26] IEC 60268-16:2020 scope: the standard does not cover the case of fluctuating noise on the Speech Transmission Index (general comments in §7.13 and §8.9.3). MWAA specification should therefore require commissioning verification under representative operating traffic in addition to an empty-building baseline, treated as verification-with-cure-period rather than an occupancy condition.

[27] 29 C.F.R. § 1910.95 (OSHA, Occupational Noise Exposure): hearing-conservation program required at 85 dB(A) eight-hour time-weighted average.